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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Nameless
Project / Solution Name	SME Manufacturing Quotation Assistant
Name (Leader)	ZHANG XINRUI
Email & Phone (Leader)	AIT2609083@xmu.edu.my & +60 1117808845
University Affiliation(s)	XIAMEN UNIVERSITY MALAYSIA
Live Prototype URL	https://zhong-xiao-zhi-zao-qi-ye-bao-jie-fu-zhu-xi-tong.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/The-Wildest-Dream/SDC-project

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The browser accesses a Replit-hosted Python Streamlit application for CSV data, inquiries, price analysis, approvals and quote records. Python services calculate costs and payment-term references and persist snapshots in PostgreSQL through SQLAlchemy and psycopg. The historical-price page sends the product, currency, unit, quantity, production cost and inquiry date to a RunPod Serverless CPU queue endpoint. The worker loads the trusted scikit-learn Ridge model and returns a historical transaction-price estimate and metadata. The app labels prediction sources and can fall back to the local artifact.
Offline training uses separate CNY and MYR synthetic groups. No GPU or runtime GenAI service is used for inference. Sales staff enter the final proposed price; simulated managers approve it. GitHub stores source and model artifacts. Replit Secrets supply database and API configuration. Chinese, English and Malay resources support the interface. Deployment version parity and production database workflows require final verification.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.13	Application runtime	PSF
Streamlit 1.66.1	Web interface	Apache-2.0
pandas 3.1.0; NumPy 2.5.4	Data processing	BSD-3-Clause
scikit-learn 1.9.1; joblib 1.6.0	Model training, inference and persistence	BSD-3-Clause (scikit-learn); BSD-3-Clause (joblib)
SQLAlchemy 2.1.6	Database access	MIT
psycopg / psycopg-binary 3.3.6	PostgreSQL driver	LGPL-3.0-or-later; bundled components retain their notices
RunPod SDK 1.12.0	Serverless worker adapter	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless REST API	Submit inference jobs at /run and poll /status/{job_id}; historical price prediction. Endpoint: https://api.runpod.ai/v2/4qbw60i5ghyhfo	Metered cloud service; sponsor-provided

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Application development and published Streamlit hosting	Replit Core. / Region not specified.
PostgreSQL through DATABASE_URL	Persistent records, inquiries, quotes and audit logs	Plan / Region not specified.
RunPod Serverless CPU	Queued remote inference using the saved Ridge model	Endpoint: 4qbw60i5ghyhfo; 25$ / Region not specified.
GitHub	Public source repository, history and Actions image build	Public repository: https://github.com/The-Wildest-Dream/SDC-project

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Historical transaction-price model; data version b6d83cf569f9 (full hash in evaluation.json)	Team train_price_model.py; StandardScaler + Ridge(alpha=1)	Self-trained offline on CPU; separate product/currency/unit groups; local and RunPod inference	Project Apache-2.0 license; underlying libraries retain their licenses

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
CNY synthetic manufacturing records	generate_data.py; sample_data/	Synthetic; 300 historical transactions plus product/customer/inventory data	Team-generated project data under repository Apache-2.0 license	N; fictional records
MYR synthetic manufacturing records	generate_myr_data.py; four MYR synthetic CSV files in sample_data/	Synthetic; 320 transactions, 4 products and 6 customers; independent MYR assumptions	Team-generated project data under repository Apache-2.0 license	N; fictional records
Combined training/evaluation inputs	train_price_model.py; models/historical_price/evaluation.json	620 synthetic transactions; 496 train / 124 test; cutoff 2026-08-13; currency groups stay separate	Team-generated project data under repository Apache-2.0 license	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent; underlying model/version not disclosed in project records	Code generation, debugging and application changes	Major
ChatGPT; model/version varies by session and is not recorded	Planning, development instructions, translation and documentation drafting	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team used staged implementation prompts and feature checks. A separate local audit installed locked dependencies on Python 3.13, passed 51 unit tests, confirmed Streamlit startup with HTTP 200 responses, and generated CNY/MYR local predictions. RunPod transport tests used mocks rather than paid remote requests. Production database approval and live endpoint parity were not verified in that audit. A draft-edit financing snapshot issue was reproduced and remains a known defect until fixed. The leader must review the final code, verify the demonstration workflow and approve this declaration before signing.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform used; official SDC credits/account allocation were used	Development, Agent assistance and application hosting
RunPod	Serverless CPU platform used; official SDC credits were used	Historical-price model inference through the declared endpoint
Other official sponsor resources	N/A	N/A

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit supports development and hosting, while RunPod serves the price model remotely. A labeled local fallback supports continuity if remote inference fails. The Python services and PostgreSQL design are portable to other compatible infrastructure. The current model runs on CPU and does not require GPU acceleration. Platform use is confirmed by the repository implementation and supplied screenshots; sponsor-funded credit redemption, billing tiers and any additional mandatory sponsor integration must be confirmed by the team.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Demo customer, product, inventory and transaction records are synthetic. Imported records, inquiries, quote snapshots and logs persist in PostgreSQL. The RunPod client sends product ID, currency, unit, quantity, production cost and inquiry date for prediction. It does not send customer names or database credentials. API credentials and DATABASE_URL are environment configuration rather than document content. Demo roles are not authenticated identities. The project does not implement an automatic data-retention or deletion schedule; the operator manages stored records. Use only synthetic data in the public demonstration.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Synthetic-data evaluation does not establish real-world price accuracy or commercial benefit. The model estimates historical prices, not optimal profit or purchase probability. It beats the median baseline in 8 of 10 groups, not all groups. Unsupported inputs can suppress predictions. RunPod queue delay or failure can trigger a labeled local fallback. Simulated roles are not production authentication. Payment histories do not establish customer integrity. Funding calculations use simplified assumptions. A known defect can leave old financing parameters after draft edits; fix and regression-test it before relying on changed cost floors. The local audit did not verify live database approvals or remote endpoint parity. Human review remains necessary.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [√] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [√] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [√] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [√] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [√] Any use of official SDC sponsor resources has been declared in Section 5.
- [√] The prototype described here is accessible to judges through the links provided in Section 1.
- [√] All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature] ZHANG XINRUI

[Date] 4th/10/2026